

考研竞赛凯哥 第6次周考

考试时间: 4月19日上午08:30~10:00

讲解时间: 4月19日上午10:00~12:00

注1: 本次周考的范围为《极限与连续》《一元微分的计算与证明》《一元积分的计算与证明》

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机;

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

1. $\lim_{x \rightarrow \infty} [\sqrt[n]{(x^2+1)(x^2+2)\cdots(x^2+n)} - x^2].$

解: $I \stackrel{x^2=\frac{1}{t}}{=} \lim_{t \rightarrow 0^+} \left[\sqrt[n]{\left(\frac{1}{t}+1\right)\left(\frac{1}{t}+2\right)\cdots\left(\frac{1}{t}+n\right)} - \frac{1}{t} \right] = \lim_{t \rightarrow 0^+} \left[\sqrt[n]{\left(\frac{1+t}{t}\right)\left(\frac{1+2t}{t}\right)\cdots\left(\frac{1+nt}{t}\right)} - \frac{1}{t} \right]$

$$= \lim_{t \rightarrow 0^+} \frac{[(1+t)(1+2t)\cdots(1+nt)]^{\frac{1}{n}} - 1}{t} = \lim_{t \rightarrow 0^+} \frac{\ln[(1+t)(1+2t)\cdots(1+nt)]^{\frac{1}{n}}}{t} = \frac{1}{n} \lim_{t \rightarrow 0^+} \frac{\ln[(1+t)(1+2t)\cdots(1+nt)]}{t}$$

$$= \frac{1}{n} \left[\lim_{t \rightarrow 0^+} \frac{\ln(1+t)}{t} + \lim_{t \rightarrow 0^+} \frac{\ln(1+2t)}{t} + \cdots + \lim_{t \rightarrow 0^+} \frac{\ln(1+nt)}{t} \right] = \frac{1}{n} (1+2+\cdots+n) = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}.$$

2. 设 $S(x) = \int_0^x |\cos t| dt$. (1) 证明: 当 $n\pi \leq x < (n+1)\pi$ 时, $2n \leq S(x) < 2(n+1)$; (2) 求 $\lim_{x \rightarrow +\infty} \frac{S(x)}{x}$.

解: (1) $S'(x) = |\cos x| \geq 0$, 故 $S(x)$ 递增, 故当 $n\pi \leq x < (n+1)\pi$ 时, $S(n\pi) \leq S(x) < S[(n+1)\pi]$.

$$S(n\pi) = \int_0^{n\pi} |\cos x| dx = n \int_0^\pi |\cos x| dx = 2n, \text{ 故 } S[(n+1)\pi] = 2(n+1), \text{ 故 } 2n \leq S(x) < 2(n+1), \text{ 证毕.}$$

(2) 由于 $\begin{cases} n\pi \leq x < (n+1)\pi \\ 2n \leq S(x) < 2(n+1) \end{cases}$, 故 $\frac{2n}{(n+1)\pi} \leq \frac{S(x)}{x} \leq \frac{2(n+1)}{n\pi}$.

当 $n\pi \leq x < (n+1)\pi$ 时, $x \rightarrow +\infty$ 和 $n \rightarrow \infty$ 是充要条件, 故在 $\frac{2n}{(n+1)\pi} \leq \frac{S(x)}{x} \leq \frac{2(n+1)}{n\pi}$ 中,

令 $x \rightarrow +\infty, n \rightarrow \infty$, 由夹逼准则可得, $\lim_{x \rightarrow +\infty} \frac{S(x)}{x} = \frac{2}{\pi}$.

3. 设 $y = y(x)$ 由方程 $e^{xy} - 1 = \int_1^{x+y} \sin^2 \frac{\pi}{4} t dt$ 确定, 求 $y''(0)$.

解: emmm, 很不想敲这道题, 纯粹的体力活.

在 $e^{xy} - 1 = \int_1^{x+y} \sin^2 \frac{\pi}{4} t dt$ 中, 令 $x=0$, 得 $0 = \int_1^y \sin^2 \frac{\pi}{4} t dt$, 故 $y=1$, 即 $y(0)=1$;

在题干等式两边对 x 求导——

$$e^{xy} [1 \cdot y + xy'] = (1+y') \sin^2 \frac{\pi}{4} (x+y) \quad (1)$$

令 $x=0, y=1$, 得 $1=(1+y')\left(\frac{\sqrt{2}}{2}\right)^2$, 解得 $y'(0)=1$.

在(1)式两边同时对 x 求导——

$$e^{xy}[1 \cdot y + xy']^2 + e^{xy}(y' + y' + xy'') = y'' \sin^2 \frac{\pi}{4}(x+y) + (1+y') \cdot 2 \sin \frac{\pi}{4}(x+y) \cos \frac{\pi}{4}(x+y) \cdot \frac{\pi}{4}(1+y')$$

在上式中令 $x=0, y=1, y'=1$, 得 $1^2 + (1+1) = y'' \left(\frac{\sqrt{2}}{2}\right)^2 + (1+1) \cdot 2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\pi}{4}(1+1)$,

即 $3 = \frac{1}{2}y'' + \pi$, 解得 $y''(0) = 6 - 2\pi$.

4. 设 $f'(e^x) = \cos x$, 求 $f(x)$.

解: 令 $e^x = t$, 故 $x = \ln t$, 故 $f'(t) = \cos \ln t$.

$$\begin{aligned} I &= \int \cos \ln x dx = x \cos \ln x - \int x(-\sin \ln x) \frac{1}{x} dx = x \cos \ln x + \int \sin \ln x dx \\ &= x \cos \ln x + x \cos \ln x - \int x(\cos \ln x) \frac{1}{x} dx \\ &= x \cos \ln x + x \cos \ln x - I. \end{aligned}$$

解得 $I = f(x) = \frac{x}{2}(\sin \ln x + \cos \ln x) + C$.

5. $\int_0^{\pi} \frac{x \sqrt{\sin^2 x - \sin^4 x}}{1 + 3 \cos^2 x} dx$.

解: 根据 $\int_0^{\pi} x f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx$ 可得——

$$\begin{aligned} \int_0^{\pi} \frac{x \sqrt{\sin^2 x - \sin^4 x}}{1 + 3 \cos^2 x} dx &= \pi \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin^2 x (1 - \sin^2 x)}}{1 + 3 \cos^2 x} dx = \pi \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{1 + 3 \cos^2 x} dx \\ &= -\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1 + 3 \cos^2 x} d(\cos^2 x) \\ &= -\frac{\pi}{2} \cdot \frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{1}{1 + 3 \cos^2 x} d(3 \cos^2 x) = -\frac{\pi}{6} \ln(1 + 3 \cos^2 x) \Big|_0^{\frac{\pi}{2}} = -\frac{\pi}{6} \ln \frac{1}{4} = \frac{\pi}{6} \ln 4 = \frac{\pi}{3} \ln 2. \end{aligned}$$

6. $\int_1^2 2x^3 \sqrt{4-x^2} dx$

解: $\int_1^2 2x^3 \sqrt{4-x^2} dx = \int_1^2 x^2 \sqrt{4-x^2} d(x^2) = \int_1^4 t \sqrt{4-t} dt$, 令 $\sqrt{4-t} = u$, 解得 $t = 4 - u^2$.

$$I = \int_{\sqrt{3}}^0 (4-u^2)u(-2u)du = 2 \int_0^{\sqrt{3}} (4-u^2)u^2 du = 2 \left[\frac{4}{3}u^3 - \frac{1}{5}u^5 \right] \Big|_0^{\sqrt{3}} = 2 \left[\frac{4}{3} \cdot 3\sqrt{3} - \frac{1}{5} \cdot 9\sqrt{3} \right] = \frac{22}{5}\sqrt{3}$$

$$7. \int_{\frac{\sqrt{3}}{3}}^1 \frac{\arctan x}{x^5} dx$$

$$\begin{aligned} \text{解: } \int \frac{\arctan x}{x^5} dx &= \int \arctan x d\left(\frac{x^{-4}}{-4}\right) = -\frac{\arctan x}{4x^4} + \frac{1}{4} \int \frac{(1+x^2)-x^2}{x^4(1+x^2)} dx \\ &= -\frac{\arctan x}{4x^4} + \frac{1}{4} \int \frac{1}{x^4} dx - \frac{1}{4} \int \frac{1}{x^2(1+x^2)} dx = -\frac{\arctan x}{4x^4} + \frac{1}{4} \cdot \frac{1}{-3} \frac{1}{x^3} - \frac{1}{4} \left(-\frac{1}{x} - \arctan x\right) + C. \end{aligned}$$

$$\text{故 } \int_{\frac{\sqrt{3}}{3}}^1 \frac{\arctan x}{x^5} dx = \left[-\frac{\arctan x}{4x^4} + \frac{1}{4} \cdot \frac{1}{-3} \frac{1}{x^3} - \frac{1}{4} \left(-\frac{1}{x} - \arctan x\right) \right] \Big|_{\frac{\sqrt{3}}{3}}^1 = \frac{1+2\pi}{6}.$$

8. 若 $x_{n+1} + \frac{4}{x_n} < 4$, $x_n > 0$, $n=1, 2, \dots$, 证明 $\{x_n\}$ 收敛, 并求 $\lim_{n \rightarrow \infty} x_n$.

解: 令 $f(x) = x + \frac{4}{x}$ ($x > 0$), $f'(x) = 1 - \frac{4}{x^2} = \frac{x^2 - 4}{x^2}$, $f(x)$ 在 $(0, 2)$ 单调递减, 在 $(2, +\infty)$ 单调递增.

故 $[f(x)]_{\min} = f(2) = 2$, 故 $f(x) \geq 2$ 恒成立, 故 $f(x_n) \geq 2$, 再结合题中不等式, 得到不等式组:

$$\begin{cases} x_{n+1} + \frac{4}{x_n} < 4 \\ x_n + \frac{4}{x_n} \geq 2 \end{cases}, \text{ 对比可得 } x_{n+1} < x_n, \text{ 又 } x_n > 0, \text{ 故由单调有界准则 } \{x_n\} \text{ 收敛, 设 } \lim_{n \rightarrow \infty} x_n = A.$$

$$\text{在上述不等式组两边取极限, 得 } \begin{cases} A + \frac{4}{A} \leq 4 \\ A + \frac{4}{A} \geq 2 \end{cases}, \text{ 夹逼得 } A + \frac{4}{A} = 4, \text{ 解得 } A = 2, \text{ 故 } \lim_{n \rightarrow \infty} x_n = 2.$$

9. 设 $f(x)$ 在 $[a, b]$ 上二阶可导, $f(a) = f(b) = f'(a) = 0$, 证明: $\exists \xi \in (a, b)$, 使得 $f''(\xi) = f(\xi)$.

解: 令 $F(x) = e^{-x}[f'(x) + f(x)]$, $F(a) = 0$, $F(b)$ 未知, 想用罗尔定理需要找到 $F(x)$ 的另一个零点.

为此, 令 $G(x) = e^x f(x)$, 由于 $G(a) = G(b) = 0$, 故由罗尔定理, 存在 $\theta \in (a, b)$, 使得 $G'(\theta) = 0$, 即 $e^\theta[f'(\theta) + f'(\theta)] = 0$, 故 $f'(\theta) + f'(\theta) = 0$, 故 $F(\theta) = 0$.

由 $F(a) = F(\theta) = 0$ 可得, 存在 $\xi \in (a, \theta) \subset (a, b)$, 使得 $F'(\xi) = 0$, 整理得 $f''(\xi) = f(\xi)$, 证毕.

10. 设 $f(x)$ 在 $[0, 2\pi]$ 上具有非负连续导数, 证明: $\left| \int_0^{2\pi} f(x) \sin nx dx \right| \leq \frac{2[f(2\pi) - f(0)]}{n} (n \in \mathbb{N}^+)$.

$$\begin{aligned} \text{解: 左边} &= \frac{1}{n} \left| \int_0^{2\pi} f(x) \sin nx d(nx) \right| = \frac{1}{n} \left| - \int_0^{2\pi} f(x) d\cos(nx) \right| = \frac{1}{n} \left| -f(x)\cos(nx) \Big|_0^{2\pi} + \int_0^{2\pi} \cos nx \cdot f'(x) dx \right| \\ &= \frac{1}{n} \left| f(0) - f(2\pi) + \int_0^{2\pi} \cos nx \cdot f'(x) dx \right| \leq \frac{1}{n} |f(0) - f(2\pi)| + \frac{1}{n} \int_0^{2\pi} f'(x) dx = \frac{2}{n} |f(2\pi) - f(0)|. \end{aligned}$$